

BRAVO Lab

Brazilian Risk, Allocation, Volatility & Options Lab

Institutional Executive Report

PRIVATE APPLICATION DRAFT - v0.2.2

A reproducible Python research prototype for Brazilian market stress, derivatives-overlay logic, active-risk diagnostics, and portfolio-governance review.

CREATED BY

Rodolfo Pereira

Research, portfolio risk, derivatives overlays, and market transmission analysis

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EXECUTIVE SUMMARY

BRAVO Lab is a portfolio-governance prototype, not a trading promise.

BRAVO Lab is designed to convert Brazilian market stress into a structured portfolio discussion. The project does not need to pretend that it can predict the market. Its value is more credible than that: it organizes evidence so a reviewer can ask better questions about exposure, downside risk, volatility income, and implementation discipline.

The central idea is simple. When Brazilian market conditions change, a portfolio should not react blindly, and it should not ignore the signal either. It needs a framework that separates ordinary noise from stress that deserves review. BRAVO Lab builds that framework by connecting market indicators, derivatives-overlay logic, active-risk diagnostics, drawdown evidence, and visual reporting inside a reproducible Python workflow.

This matters because most beginner quant projects stop at charts. BRAVO Lab tries to move one step closer to professional work. It turns analysis into decision material. It asks what a portfolio team would actually need to know before changing exposure: what the signal says, what strategy options exist, what each option gives up, what remains unvalidated, and which assumptions still need to be challenged before the system could be used in a live setting.

For a Nu/Nubank reviewer, the project should be positioned as evidence of applied research capability. It shows initiative, Python execution, Brazilian market awareness, derivatives reasoning, and the ability to translate technical work into a memo that a non-technical reviewer can understand. The right claim is not that BRAVO Lab is finished. The right claim is that it is a serious prototype with a clear path toward stronger validation.

WHAT BRAVO LAB IS

A clean research workflow around one practical question: what should the portfolio do when stress changes?

BRAVO Lab is a Python-based research project focused on Brazilian market risk, B3 exposure, volatility regimes, derivatives overlays, and portfolio-review logic. It is built around a practical investment-governance question: when stress rises, should a portfolio remain passive, collect option premium, buy protection, or wait for stronger confirmation?

That question matters because each answer carries a cost. Passive beta keeps full participation but leaves the portfolio exposed when stress persists. A covered-call overlay can harvest income, but it sells part of the recovery. A collar can reduce downside risk, but protection has a cost and may limit upside. A stress-aware policy can introduce discipline, but only if it is validated against false positives, transaction costs, liquidity, and live option-chain constraints.

The project is therefore not only a model. It is a decision architecture. The purpose of the code is to produce evidence. The purpose of the evidence is to support a better portfolio conversation. The purpose of the report is to make that conversation clear enough for a reviewer to inspect, challenge, and improve.

WHY IT MATTERS FOR NU / NUBANK

The strongest asset is the workflow behind the report.

BRAVO Lab should be presented to Nu/Nubank as evidence that you can build a bridge between technical research and financial decision-making. In real teams, the challenge is rarely just running code. The challenge is producing analysis that can be trusted, explained, questioned, and improved.

The project demonstrates several relevant capabilities at once. It shows that you can structure a Python research pipeline, handle processed data, generate figures, organize outputs, think in terms of risk and portfolio trade-offs, and communicate the result in a format that is easier for a recruiter or risk professional to evaluate.

The Brazilian focus makes the project more distinctive. It is not a generic S&P; 500 notebook copied from an online tutorial. It is built around B3 exposure, local stress behavior, derivatives overlays, drawdown discipline, and the type of applied-market reasoning that is closer to portfolio analytics than to academic abstraction.

The correct application message is precise: BRAVO Lab is a reproducible research prototype that turns Brazilian market stress, derivatives-overlay logic, and active-risk diagnostics into portfolio-governance evidence. That wording is stronger than claiming alpha. It is more professional, more defensible, and more aligned with the way real risk teams evaluate work.

PORTFOLIO GOVERNANCE PROBLEM

The real issue is not volatility. It is what the portfolio is allowed to do when volatility changes.

A weak report says that volatility increased. A stronger report asks what that change should mean for portfolio behavior. Should the portfolio keep full exposure? Should it monetize volatility through a covered-call structure? Should it pay for protection through a collar? Should it wait because the signal is not persistent enough?

This is where BRAVO Lab becomes more useful. It reframes the problem from prediction to governance. A model does not have to remove judgment. A serious model should organize judgment. It should make the assumptions visible, the trade-offs explicit, and the decision easier to challenge.

The most important professional question is not whether one strategy wins in a backtest. The important question is whether the chosen overlay is defensible under a specific regime. If stress is temporary, action may create noise. If stress is persistent, inaction may be expensive. If protection is too costly, the hedge may destroy the return profile. If income is harvested too aggressively, the portfolio may sell recovery exactly when it needs participation.

METHOD ARCHITECTURE

From market evidence to decision material.

The BRAVO workflow begins with market data and processed indicators. These inputs feed stress diagnostics, strategy comparison, drawdown analysis, policy-selection logic, and executive reporting. The point is not to create a decorative dashboard. The point is to create a chain of evidence that can be traced from raw data to portfolio interpretation.

A clean architecture matters because it protects the project from becoming a loose collection of charts. Each component has a role. The stress indicator identifies pressure. The strategy comparison shows alternative responses. The drawdown profile tests behavior under loss. The policy layer connects regimes to action. The report turns the outputs into a narrative that a reviewer can understand without opening every file first.

This is why reproducibility is central. A serious reviewer should be able to move from the PDF to the repository and find the supporting materials: processed CSV files, figures, technical reports, and scripts. The PDF should be the readable front door, not the only evidence.

Layer	Institutional role
Data and processed files	Create the evidence base for stress indicators, strategy behavior, and reporting outputs.
Stress diagnostics	Translate market pressure into a reviewable signal instead of a vague market feeling.
Strategy comparison	Make the trade-off between passive beta, income, protection, and switching visible.
Drawdown evidence	Test whether the overlay improves behavior when the portfolio is under pressure.
Policy logic	Connect stress states to possible actions while remaining open to validation and challenge.
Executive report	Translate the technical work into decision language for recruiters, risk reviewers, or portfolio teams.

STRATEGY COMPARISON

The point is not a magic strategy. The point is a disciplined comparison.

A passive strategy is clean and cheap, but it accepts the full drawdown profile of the underlying exposure. A covered-call overlay can create income, but the income is not free; it is paid for by giving up part of the upside. A collar can reduce left-tail pain, but it introduces hedge cost and changes the participation profile. A regime-aware policy can create discipline, but it must prove that its switching behavior is not just overfitting dressed as intelligence.

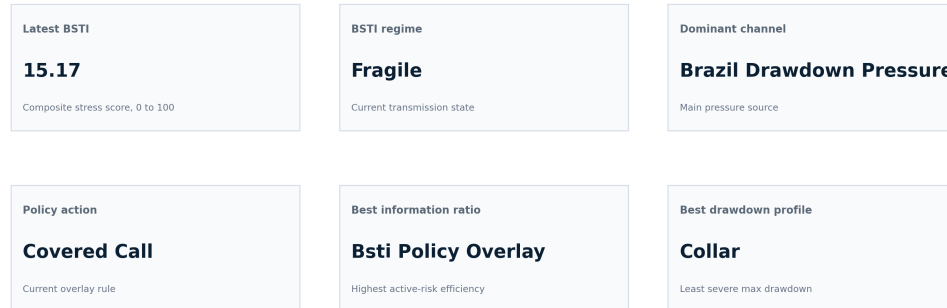
This is why BRAVO Lab should be read as a governance tool. It does not need to say that one strategy is always superior. In fact, that would be a weaker claim. The stronger argument is that each strategy has a role under specific conditions, and the model helps make those conditions easier to review.

Strategy	Professional interpretation
Passive beta	Best when participation matters more than protection and the stress signal is not strong enough to justify intervention.
Covered call	Useful when volatility income is attractive and the portfolio can accept limited upside in exchange for premium.
Collar	Useful when drawdown discipline matters more than full participation and the cost of protection is acceptable.
BSTI policy	Useful as a review rule, but only after validation against false positives, trading friction, liquidity, and real option-chain behavior.

Evidence dashboard

BRAVO Lab Executive Risk Dashboard

Brazilian risk, allocation, volatility, options, and stress-transmission intelligence



Source: BRAVO Lab processed outputs. Dashboard is generated automatically from reproducible CSV evidence files.

Figure: 00_executive_risk_dashboard.png

The dashboard should be treated as the front page of the analytical evidence, not as a substitute for the model. Its job is to summarize the current state quickly: the stress signal, regime label, policy output, and risk profile. A reviewer should be able to look at it and immediately ask whether the current evidence justifies a change in Brazil equity exposure.

Cumulative performance evidence

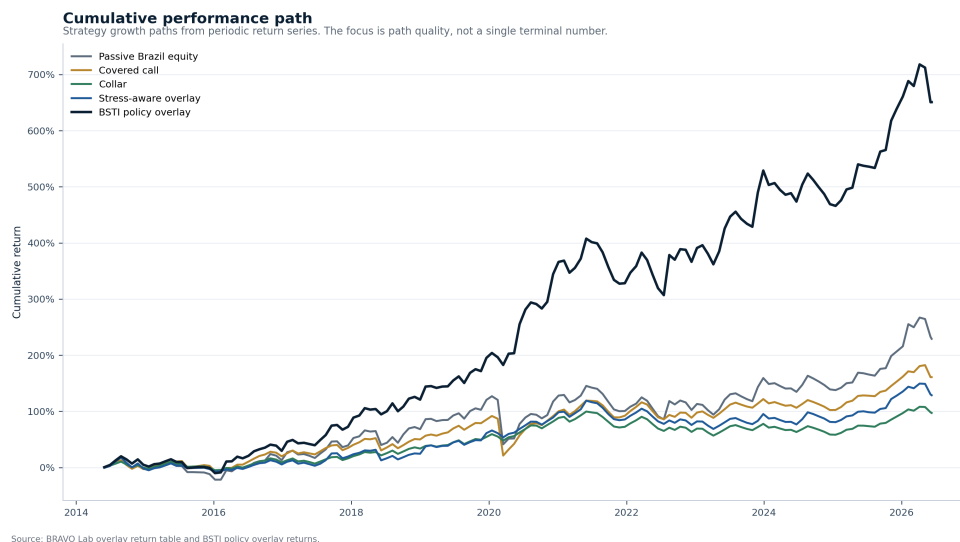


Figure: 01_cumulative_performance.png

The cumulative performance path is useful only when it is read with caution. A rising line is not enough. A portfolio reviewer needs to know what risk was taken to produce the path, whether the result depends on a few periods, whether the overlay sacrifices recovery, and whether implementation costs would erode the apparent advantage.

Drawdown and risk-control evidence

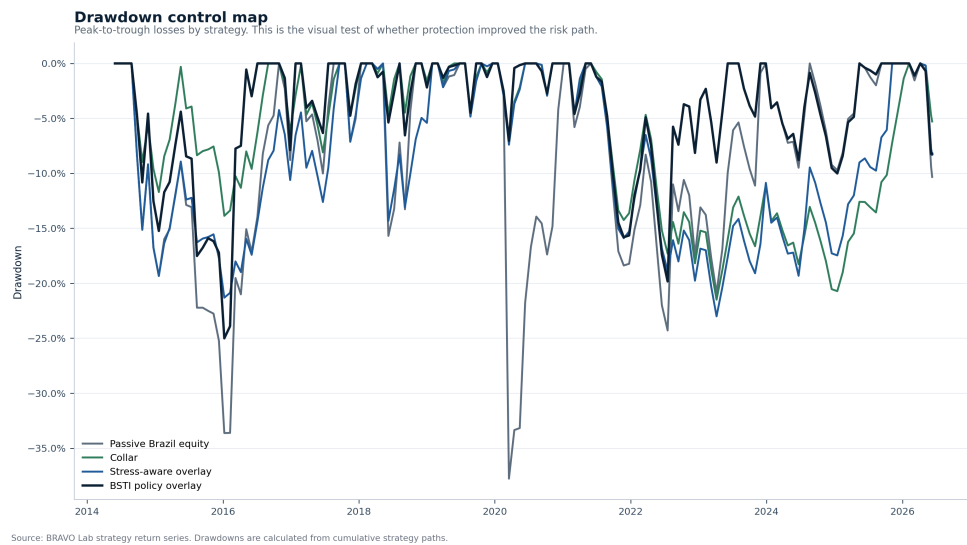


Figure: 02_drawdown_profile.png

Drawdown evidence is more important than it looks. It reveals what the strategy does when the portfolio is under stress. If an overlay cannot improve the pain profile, it has little reason to exist. If it improves drawdown but destroys participation, that trade-off must also be visible. The professional question is not whether the line looks better. The question is whether the improvement is worth the cost.

BSTI signal logic

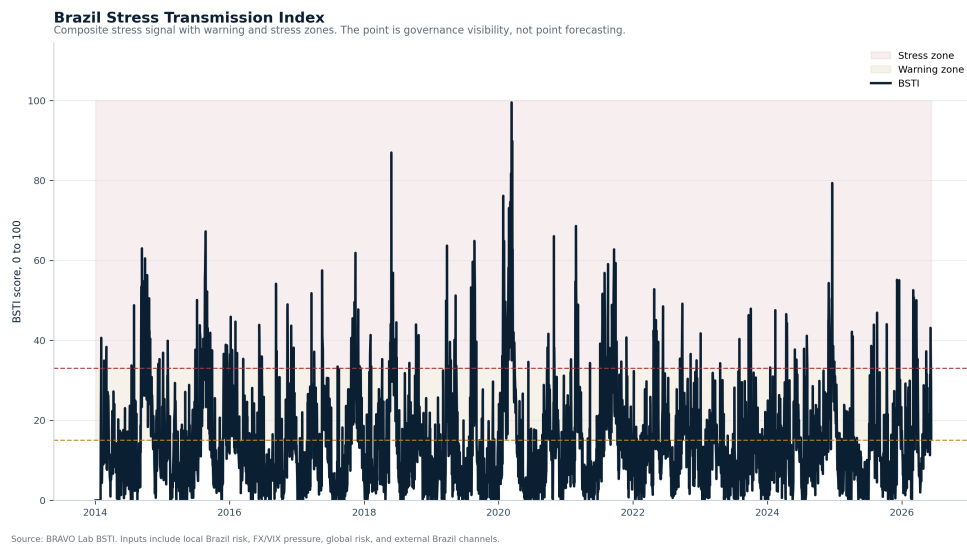


Figure: 03_bsti_signal.png

The Brazil Stress Transmission Index should be presented as a review trigger, not as a prediction engine. Its purpose is to help separate ordinary volatility from conditions that deserve closer examination. The key question is threshold discipline: when does the signal justify observation, hedge review, income overlay, downside protection, or no action at all?

Policy-selection logic

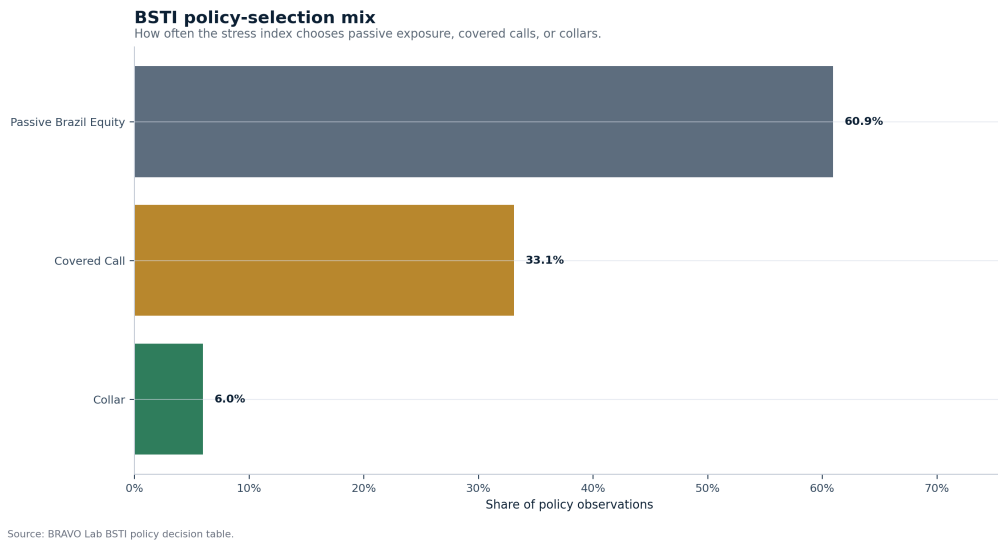


Figure: 04_bsti_policy_selection_mix.png

The policy mix matters because it shows whether the model is mostly passive, mostly defensive, income-oriented, or too active. A policy rule that changes behavior too often can become a hidden source of cost. A rule that never changes behavior is not a rule. The value sits in the middle: disciplined action only when the evidence deserves it.

Regime persistence evidence

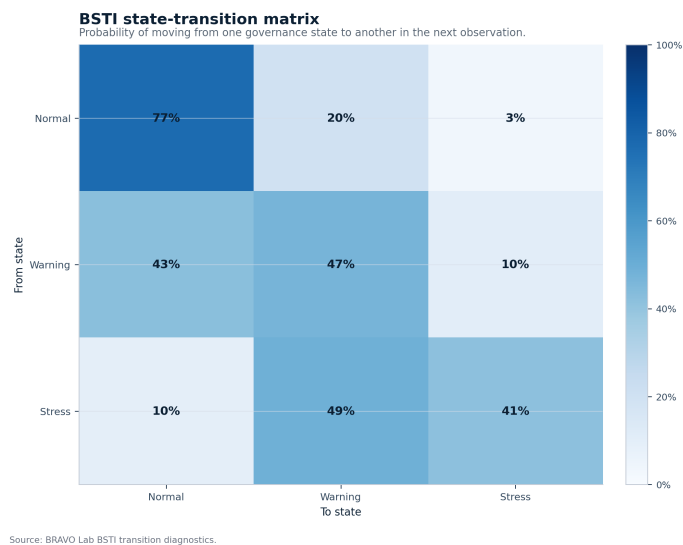


Figure: 06_bsti_transition_matrix.png

Regime persistence is central to action. A temporary warning state may not justify an overlay change. A state that tends to remain fragile or escalate into stress deserves more attention. This evidence helps convert the stress signal from a single observation into a question about persistence, transition, and the probability of unnecessary intervention.

Calibration evidence

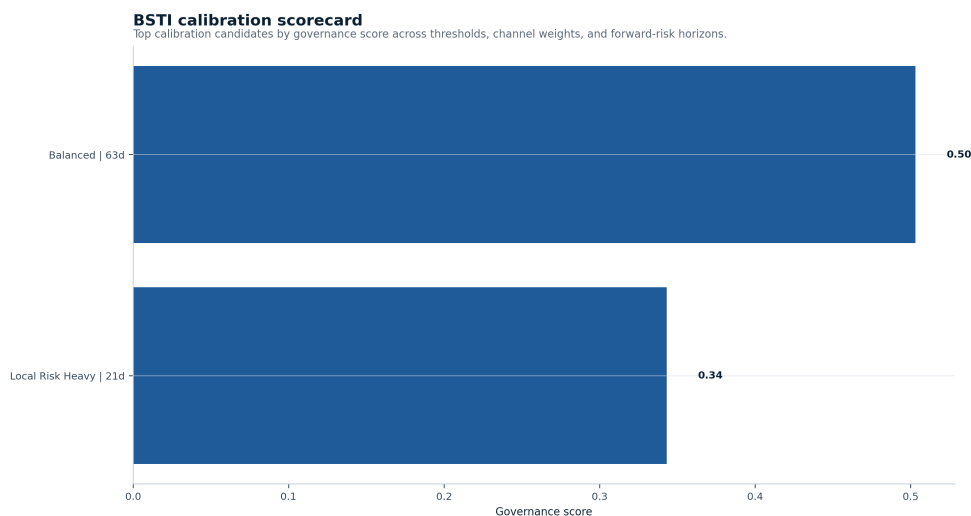


Figure: 07_bsti_calibration_scores.png

Calibration is where the prototype becomes auditable. A stress rule depends on thresholds, definitions, and horizon choices. If a calibration works only under one narrow setting, the signal is fragile. If it remains useful across reasonable settings, it becomes more credible. This page should be read as part of the validation layer, not as a final proof.

VALIDATION LIMITS

The most credible page is the one that admits what still needs to be tested.

BRAVO Lab should be explicit about its current limits. The prototype is useful as evidence of applied research infrastructure, but the derivatives layer still requires observed B3 option-chain validation. Synthetic option assumptions are acceptable for early modeling, but they cannot support a final implementation claim.

The next validation layer should include real strikes, maturities, bid-ask spreads, liquidity filters, transaction costs, taxes, roll rules, and exercise assumptions. Without those components, the project should remain framed as decision-support research rather than an investable strategy.

This honesty is not a weakness. It is exactly what makes the project more professional. A reviewer can trust a prototype more when the author is clear about what is proven, what is simulated, what is assumed, and what still needs to be validated.

DEVELOPMENT ROADMAP

The next build should increase trust, not complexity.

The next version of BRAVO Lab should not add complexity for its own sake. Every new layer should answer one professional question: does it make the portfolio decision more transparent, more testable,

or more realistic?

The first priority is real B3 option-chain validation. The second is implementation friction: transaction costs, taxes, spreads, roll mechanics, and liquidity. The third is a stronger volatility-dependence layer through MTV-GARCH, DCC, or related models. The fourth is stress compression through CCA or Ridge CCA. The fifth is multiscale analysis through Airy or Generalized Morse Wavelets, especially if the project evolves toward high-dimensional stress transmission.

This roadmap matters because it gives the project a serious future without overselling the present. It tells the reviewer that the prototype is not random. It has a logical path from application evidence to a more robust research system.

Next layer	Why it matters
Real B3 option chains	Moves the overlay logic from synthetic assumptions toward observed market implementation.
Transaction costs	Tests whether apparent strategy benefits survive friction, taxes, spreads, and roll mechanics.
Volatility dependence	Improves the risk model by connecting stress to covariance and changing market relationships.
CCA stress compression	Helps identify which market blocks transmit pressure most clearly into the portfolio.
Wavelet layer	Separates short-term noise from persistent stress across multiple horizons.
Nu application note	Translates the project into simple recruiter language without weakening the technical foundation.

DEEP SWOT

The project becomes stronger when the limitations are part of the argument.

The main strength of BRAVO Lab is that it already has an end-to-end structure. It is not only an idea. It has scripts, figures, processed outputs, reports, and a clear decision problem. It also has a distinctive Brazilian-market angle, which helps separate it from generic portfolio projects.

The main weakness is validation depth. The derivatives layer still needs real B3 option data. The project also needs more explicit treatment of trading costs, taxes, liquidity, and turnover. Those weaknesses are manageable because they are clear. What would be more dangerous is pretending they do not exist.

The opportunity is to turn BRAVO into a strong evidence portfolio for quant, risk, and portfolio analytics applications. The threat is overclaiming. If the project is described as a market-prediction engine, it becomes vulnerable. If it is described as a reproducible decision-support prototype, it becomes much more credible.

SWOT	Clean institutional read
Strengths	Applied Python workflow, Brazil-specific market lens, derivatives-overlay reasoning, reproducible outputs, and executive communication.

Weaknesses	Real option-chain validation, transaction costs, liquidity filters, taxes, and live implementation constraints still need deeper work.
Opportunities	Can evolve into a serious portfolio analytics project with MTV-GARCH, CCA, wavelets, and observed B3 derivatives data.
Threats	Loses credibility if framed as prediction, investment advice, or a finished production engine.

EVIDENCE MAP

A serious prototype should be easy to inspect.

The PDF should guide the reviewer toward the project evidence. It should not replace the repository. A strong institutional report makes the logic readable, then points back to the files that support the argument.

The core evidence includes the front-office memo, the technical baseline report, the generated figures, the processed CSV outputs, the PDF generator, and the Nu/Nubank application note. Together, these files show that BRAVO Lab is not merely a visual mockup. It is a reproducible research workflow.

Evidence	Project location
Executive memo	reports/front_office_memo.md
Technical report	reports/baseline_report.md
Figures	reports/figures/
Processed outputs	data/processed/
PDF generator	scripts/generate_institutional_pdf.py
Application note	reports/nu_asset_application_note.md

FINAL POSITIONING

The clean claim is stronger than the dramatic one.

BRAVO Lab should be presented as a reproducible Python research prototype for Brazilian market stress, derivatives overlays, active-risk diagnostics, and portfolio-governance review. That is a strong claim because it is specific, testable, and professionally defensible.

The project does not need exaggerated language. It needs clarity. It shows that you can build, organize, explain, and improve a financial research system. For an application, that is the real value.

BRAVO Lab

From Brazilian market stress to portfolio-governance review.

Research prototype - Private application evidence - Not investment advice

Created by Rodolfo Pereira